



FOR OFFICIAL USE

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**X816/76/01**

**Computing Science**

Duration — 2 hours



Fill in these boxes and read what is printed below.

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Total marks — 80

**SECTION 1 — Software design and development and Computer systems — 55 marks**

Attempt ALL questions.

Attempt EITHER Section 2 OR Section 3

**SECTION 2 — Database design and development — 25 marks**

**SECTION 3 — Web design and development — 25 marks**

Write your answers clearly in the spaces provided in this booklet. Additional space for answers is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.



SECTION 1 — SOFTWARE DESIGN AND DEVELOPMENT AND COMPUTER SYSTEMS  
— 55 marks

Attempt ALL questions

1. Two's complement can be used to represent positive and negative integers.

- (a) Convert the denary number -9 into 8-bit two's complement.

1

- (b) State the range of denary values that can be represented using 8-bit two's complement.

2

2. Increasing clock speed is one method of improving processor performance.

- (a) State one other method of improving processor performance.

1

- (b) Explain how your answer to part (a) improves performance.

1



3. Describe an intelligent system used on a car journey that is beneficial for the environment.

Justify your answer.

2

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4. Write the binary number  $-0.0011$  using floating-point representation.  
There are 16 bits for the mantissa (including the sign bit) and 8 bits for the exponent.

3

*Space for working*

| sign | mantissa | exponent |
|------|----------|----------|
|      |          |          |

5. An online bank uses digital signatures when sending financial documents.  
Describe the purpose of a digital signature when sending documents.

2

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6. A function is used to ensure that a user's input for their date of birth contains exactly six numbers.

For example, someone with a date of birth of 31<sup>st</sup> May 2003 would be entered as 310503.

The first version of the code for this function is shown below.

```
...
Line 235 FUNCTION validateDob (STRING dob) RETURN BOOLEAN
Line 236     DECLARE valid INITIALLY false
Line 237     DECLARE lengthCheck INITIALLY false
Line 238     DECLARE countValid INITIALLY 0
Line 239     DECLARE thisValue INITIALLY 0

Line 240     IF length(dob) = 6 THEN
Line 241         SET lengthCheck TO true
Line 242     END IF

Line 243     FOR EACH character FROM dob
Line 244         SET thisValue TO <ASCII value of character>
Line 245         IF thisValue >= 48 AND thisValue <=57 THEN
Line 246             SET countValid TO countValid + 1
Line 247         END IF
Line 248     END FOR EACH

Line 249     IF _____ AND _____ THEN
Line 250         SET valid TO true
Line 251     END IF
Line 252     RETURN valid
Line 253 END FUNCTION
```

- (a) Complete line 249 in the space below.

2

- (b) Explain why this version of the function is inefficient.

1

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7. An athletics competition takes place between different clubs. The competition includes four throwing events which are shot put, discus, hammer and javelin.

The result for each competitor is stored in a csv file. The file stores the competitor's name, club, event name and distance thrown in metres.

A program is required to read the data from the csv file and then process it.

A sample of the data is shown below.

F Dean,Rothesay Rovers,Discus,58.04

J Smith,Hawick Harriers,Shot Put,17.23

K Singh,Rothesay Rovers,Javelin,71.75

...

- (a) The data from the file is imported into an array of records.

- (i) Using a programming language of your choice, define a suitable data structure to store the data.

2

- (ii) The csv file contains 800 results.

Using a programming language of your choice, declare a variable that can store the data for the 800 results. Your answer should include the data structure from part (i).

2

[Turn over



\* X 8 1 6 7 6 0 1 0 5 \*

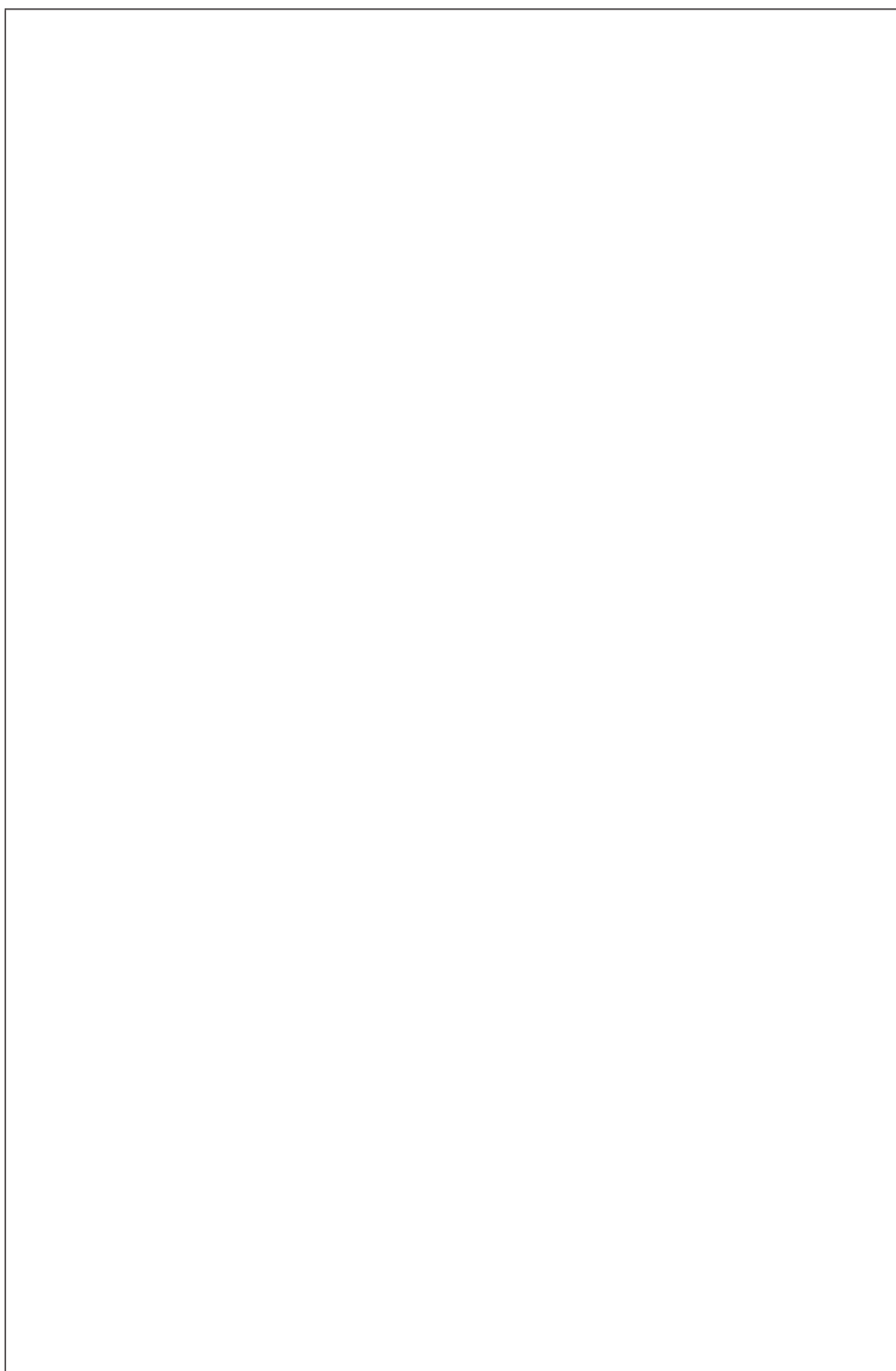
**7. (continued)**

- (b) In order to qualify for the javelin final, a competitor must throw a distance of 70 metres or more.

The event organisers would like the program to create a file with the name and club of all the competitors who have qualified for the javelin final.

Using a recognised design technique, design an algorithm that would create this file.

**5**




**7. (continued)**

- (c) The event organisers would like to be able to display the distance thrown by a competitor in a particular event.

Using a programming language of your choice, write code that

- asks for the name of a competitor and the event
- displays the competitor name, distance and event, for example 'J Smith threw 17.23 in the Shot Put' or displays the message 'Competitor not found'.

Your answer should make use of the data structure defined in part (a).

**6**



8. An app uses three selected letters from a user's password to help identify a user when logging in. An example is shown below.

Please enter the letters requested from your password.  
For example, if your password was "London" and you are asked for letters 1, 3 and 5 then you should enter L, n, o

| Letter 1 | Letter 3 | Letter 5 |
|----------|----------|----------|
| *        | *        | *        |

The top-level design for this part of the program is shown below.

1. Ask user for username and find password
2. Generate three random positions within length of password
3. Ask user for letters at generated random positions and check for a match

- (a) Complete the table below to show the missing data flow in steps 2 and 3. 2

| Step | IN/OUT | Data Flow                    |
|------|--------|------------------------------|
| 1    | IN     |                              |
|      | OUT    | password                     |
| 2    | IN     |                              |
|      | OUT    | randPos1, randPos2, randPos3 |
| 3    | IN     |                              |
|      | OUT    | valid                        |

- (b) By considering the data flow in part (a), explain why it would be appropriate to make use of a function rather than a procedure for step 3. 1

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## 8. (continued)

- (c) Explain why the programmer refers to the data flow in the top-level design when implementing the code for the program.

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[Turn over



\* X 8 1 6 7 6 0 1 0 9 \*

8. (continued)

(d) The code for step 2 is shown below.

```
...
Line 45 PROCEDURE getLetters (STRING word, INTEGER first,
                             INTEGER second, INTEGER third)
Line 46   DECLARE wordLength AS INTEGER INITIALLY
                             length(word)
Line 47   SET first TO <random number from 1 to wordLength>
Line 48   SET second TO <random number from 1 to wordLength>
Line 49   SET third TO <random number from 1 to wordLength>
Line 50 END PROCEDURE
...
Line 60 getLetters(password, randPos1, randPos2, randPos3)
...
```

(i) Formal and actual parameters are used in the code above.

Identify one formal parameter and its **associated** actual parameter. 2

Formal parameter \_\_\_\_\_

Actual parameter \_\_\_\_\_

(ii) State the scope of the variable `wordLength`. 1

\_\_\_\_\_  
\_\_\_\_\_

(iii) During testing it was found that users were asked for duplicate letters, for example asking for letters 1, 1 and 7 of their password.

Describe how the procedure could be altered to ensure three different letters are requested from the user. 2

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
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\* X 8 1 6 7 6 0 1 1 0 \*

8. (continued)

(e) Explain why the use of procedures improves the efficiency of code.

2

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[Turn over



\* X 8 1 6 7 6 0 1 1 1 \*

9. JustDine is developing a smartphone app to allow a user to identify restaurants based on the type of food. Restaurants register with JustDine. The app will store the name, city, type of food and rating for each restaurant.

Features of the app:

- Users can enter the type of food and city
- Displays the number of matching restaurants
- Displays the name and rating of the highest rated matching restaurant.

The user interface for the app is shown below.



- (a) During analysis the inputs, processes and outputs were identified. Describe two processes.

2

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## 9. (continued)

(b) Details for registered restaurants are stored in four parallel 1-D arrays.

- `restaurantName`
- `foodType`
- `city`
- `rating`

Using a recognised design technique, design an algorithm that would count the number of restaurants that match the user's food type and city.

4

[Turn over



\* X 8 1 6 7 6 0 1 1 3 \*

- (c) The following function is to be used to find the highest rating for a restaurant.

During testing it is found that this code contains an error.

```

...
Line 26 FUNCTION findMax (ARRAY OF REAL list) RETURNS
      REAL
Line 27   DECLARE upper INITIALLY length(list) -1
Line 28   DECLARE highest INITIALLY list[0]
Line 29   FOR index FROM 1 TO upper DO
Line 30       IF highest > list[index] THEN
Line 31           SET highest TO list[index]
Line 32       END IF
Line 33   END FOR
Line 34   RETURN highest
Line 35 END FUNCTION
...

```

A trace table is used when the function is tested with the following values [4.51, 4.12, 4.99].

| Line Number | index | list [index] | highest | highest > list [index] |
|-------------|-------|--------------|---------|------------------------|
| ...         |       |              |         |                        |
| 28          |       |              | 4.51    |                        |
| 29          | 1     |              |         |                        |
| 30          |       | 4.12         | 4.51    | true                   |
| 31          |       |              | A       |                        |
| 29          | 2     |              |         |                        |
| 30          |       | B            | 4.12    | C                      |
| 34          |       |              | 4.12    |                        |

- (i) Identify the missing values A, B and C.

3

A \_\_\_\_\_

B \_\_\_\_\_

C \_\_\_\_\_

- (ii) Describe how a breakpoint could be used to check the values in the trace table.

2

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_



\* X 8 1 6 7 6 0 1 1 4 \*

9. (c) (continued)

(iii) Describe how the error in this code can be corrected.

1

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(d) The function `findMax` in part (c) is used to assign the highest rating from the array `rating` to the variable `myHighest`.

Using a programming language of your choice, write the line of code that assigns the highest rating to `myHighest`.

2

[END OF SECTION 1]

[Turn over



**SECTION 2 — DATABASE DESIGN AND DEVELOPMENT — 25 marks**

**Attempt ALL questions**

10. A relational database is used to store data about the products that a shop sells. The database has the following tables.

| Product                                                                                | Category                                                  | Courier                                                           | Manufacturer                                                                      |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <u>productID</u><br>name<br>description<br>price<br>categoryName*<br>manufacturerName* | <u>categoryName</u><br>categoryDescrip<br>currentDiscount | <u>courierID</u><br>courierName<br>courierAddress<br>courierTelNo | <u>manufacturerName</u><br>manufacturerAddress<br>manufacturerTelNo<br>courierID* |

Draw an entity-relationship diagram to show the relationships that exist in this database.

Your answer should show the entity names and cardinality.

Attributes are not required on the diagram.

**3**



11. A database table is shown below.

| Film   |                         |             |         |        |
|--------|-------------------------|-------------|---------|--------|
| filmID | title                   | releaseDate | runTime | rating |
| SC1    | Spider craze            | 01/12/2018  | 117     | 12A    |
| WM1    | Waterman                | 21/12/2018  | 143     | PG     |
| IL1    | Isn't it love           | 13/02/2019  | 89      | 12A    |
| BW1    | Battle of the world     | 14/02/2019  | 122     | 12A    |
| HT1    | How to train your puppy | 22/02/2019  | 104     | PG     |
| IC2    | Ice cold                | 22/02/2019  | 119     | 15     |
| MB1    | Miss Becca              | 03/03/2019  | 104     | 15     |
| TA1    | The afterlife           | 05/03/2019  | 109     | 15     |
| CD1    | Captain Delsie          | 08/03/2019  | 124     | 12A    |
| GB1    | Green band              | 12/03/2019  | 130     | 12A    |

Complete the table below showing the expected output from the following SQL statement.

3

```
SELECT rating, MAX(runTime) AS [Longest movie]
FROM Film
GROUP BY rating
ORDER BY MAX(runTime) DESC;
```

| rating | Longest movie |
|--------|---------------|
|        |               |
|        |               |
|        |               |

[Turn over



\* X 8 1 6 7 6 0 1 1 7 \*

12. A shop uses a relational database to keep track of stock. One of the database tables is shown below.

| Snack     |                        |            |        |       |       |
|-----------|------------------------|------------|--------|-------|-------|
| productID | productName            | type       | weight | price | stock |
| 001       | Baked Cheese Snacks    | Single     | 37.5   | 0.65  | 565   |
| 002       | Sweet Chilli Flakes    | Single     | 28     | 0.65  | 300   |
| 003       | Chicken Crisps         | Single     | 32.5   | 0.55  | 240   |
| 004       | Ready Salted Crisps    | Single     | 30     | 0.58  | 654   |
| 005       | Cheese Straws          | Single     | 12     | 0.70  | 120   |
| 006       | Tomato Puffs           | Single     | 20     | 0.70  | 400   |
| 007       | Prawn Cocktail Crisps  | Multi-bag  | 78.6   | 1.00  | 335   |
| 008       | Wheat Crunch Crisps    | Multi-bag  | 138    | 1.00  | 356   |
| 009       | Assorted Flat Crisps   | Multi-bag  | 153    | 1.50  | 545   |
| 010       | Variety Pack           | Multi-bag  | 150    | 0.56  | 678   |
| 011       | Salted Pack            | Multi-bag  | 150    | 1.30  | 614   |
| 012       | Salted Pretzels        | Family bag | 80     | 0.45  | 450   |
| 013       | Tortilla Chips         | Family bag | 200    | 0.46  | 456   |
| 014       | Aberdeen Angus Crisps  | Family bag | 225    | 1.89  | 684   |
| 015       | Salt and Pepper Shakes | Family bag | 150    | 0.99  | 600   |



\* X 8 1 6 7 6 0 1 1 8 \*

## 12. (continued)

- (a) Design a query to display the most expensive multi-bag and family bag as shown below.

4

| type       | Most Expensive Item |
|------------|---------------------|
| Multi-bag  | 1.50                |
| Family bag | 1.89                |

|                             |       |
|-----------------------------|-------|
| Field(s) and calculation(s) |       |
| Tables(s) and query         | Snack |
| Search criteria             |       |
| Grouping                    |       |
| Sort order                  |       |

- (b) The manager wants to half the price of all products with stock levels of more than 500.

Write the SQL statement that would make these changes.

3

[Turn over



\* X 8 1 6 7 6 0 1 1 9 \*

12. (continued)

- (c) A query is written to display the total amount of stock for each type of snack.

The expected ordered output is shown below.

| type       | Total Stock |
|------------|-------------|
| Multi-bag  | 2528        |
| Single     | 2279        |
| Family bag | 2190        |

When the SQL statement below is tested the actual output does not match the expected output.

```
SELECT type, COUNT(stock)
FROM Snack
GROUP BY type;
```

Re-write the SQL statement to produce the expected output.

3

- (d) Write the SQL statement to remove all records of snacks that include 'Salt' in any part of their product name.

2



\* X 8 1 6 7 6 0 1 2 0 \*

13. A holiday rental shop rents a range of items to customers. It uses a relational database consisting of three linked tables storing data on items, customers and rentals.

When initially designing the `Rental` table for this database, the primary key was designed using a compound key of `customerID` and `itemID`.

| Rental             |
|--------------------|
| <u>customerID*</u> |
| <u>itemID*</u>     |
| startDate          |
| noOfDays           |

An example of the data to be stored in the `Rental` table is shown below.

| Rental     |        |            |          |
|------------|--------|------------|----------|
| customerID | itemID | startDate  | noOfDays |
| AM974      | CB001  | 26/06/2021 | 14       |
| AM974      | KB001  | 30/06/2021 | 3        |
| DR734      | MS003  | 05/07/2021 | 7        |
| AM974      | RB002  | 30/06/2021 | 3        |
| JS003      | KB002  | 04/07/2021 | 2        |
| JS003      | KB002  | 10/07/2021 | 1        |
| JS003      | RB001  | 04/07/2021 | 2        |
| KW001      | WH001  | 10/07/2021 | 14       |
| RB435      | MS003  | 15/07/2021 | 14       |
| SP234      | CB001  | 02/08/2021 | 7        |
| DR734      | KB001  | 06/07/2021 | 6        |
| DR734      | IR001  | 07/07/2021 | 2        |
| ...        | ...    | ...        | ...      |

- (a) Explain why a compound key of `customerID` and `itemID` would mean that the table is not fit for purpose.

1

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## 13. (continued)

- (b) The `Rental` table was re-designed with a primary key called `rentalID`.  
Extracts from the three tables are shown below.

| Rental   |            |        |            |          |
|----------|------------|--------|------------|----------|
| rentalID | customerID | itemID | startDate  | noOfDays |
| REN14    | AM232      | EK001  | 01/08/2021 | 14       |
| REN1     | AM974      | CB001  | 26/06/2021 | 14       |
| REN2     | AM974      | KB001  | 30/06/2021 | 3        |
| ...      | ...        | ...    | ...        | ...      |

| Customer   |           |         |           |           |
|------------|-----------|---------|-----------|-----------|
| customerID | firstName | surname | hotel     | resort    |
| AM232      | Amy       | Moore   | Esmerelda | Caleta    |
| AM974      | Amy       | Moore   | Labranda  | Jandia    |
| DR734      | Darcia    | Rycroft | Arenas    | Corralejo |
| ...        | ....      | ...     | ...       | ...       |

| Item   |                  |           |                  |
|--------|------------------|-----------|------------------|
| itemID | itemType         | itemModel | dailyRentalPrice |
| CB001  | Children's buggy | Basic     | £6.00            |
| CB002  | Children's buggy | Mid       | £7.00            |
| CB003  | Children's buggy | Deluxe    | £9.00            |
| EK001  | Kettle           | Mid       | £4.00            |
| IR001  | Iron             | Mid       | £3.00            |
| KB001  | Kid's bike       | Basic     | £14.00           |
| ...    | ...              | ...       | ...              |



13. (b) (continued)

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A list is required to show details of all rentals with a start date in July 2021. The list should display the calculated rental cost for each rental, as shown below.

| customerID | itemType          | startDate  | Rental Cost |
|------------|-------------------|------------|-------------|
| JS001      | Manual wheelchair | 02/07/2021 | £32.00      |
| JS003      | Road bike         | 04/07/2021 | £40.00      |
| JS003      | Kid's bike        | 04/07/2021 | £30.00      |
| DR734      | Mobility scooter  | 05/07/2021 | £126.00     |
| DR734      | Kid's bike        | 06/07/2021 | £84.00      |
| DR734      | Iron              | 07/07/2021 | £6.00       |
| AM974      | Road bike         | 08/07/2021 | £42.00      |
| KW001      | Manual wheelchair | 10/07/2021 | £112.00     |
| JS003      | Kid's bike        | 10/07/2021 | £15.00      |
| KW001      | Kettle            | 12/07/2021 | £48.00      |
| RB435      | Mobility scooter  | 15/07/2021 | £252.00     |

Design a query to produce this sorted output.

4

|                             |  |
|-----------------------------|--|
| Field(s) and calculation(s) |  |
| Tables(s) and query         |  |
| Search criteria             |  |
| Grouping                    |  |
| Sort order                  |  |



\* X 8 1 6 7 6 0 1 2 3 \*

## 13. (continued)

- (c) A query was created to display the highest daily rental price. This query was saved as 'MaxRental'.

| Max daily rental |
|------------------|
| 23.00            |

Using the 'MaxRental' query, complete the SQL statement below to display the item type and item model of any item matching this highest price.

2

```
SELECT itemType, itemModel
```



\* X 8 1 6 7 6 0 1 2 4 \*

[Turn over for SECTION 3

DO NOT WRITE ON THIS PAGE



### SECTION 3 — WEB DESIGN AND DEVELOPMENT — 25 marks

Attempt ALL questions

14. The HTML code and JavaScript function below changes the colour of text when the mouse moves over it.

```
<script>
    function displayRed(my_text)
    {my_text.style.color='red';}
</script>
```

```
<p onmouseover="displayRed(this)">Sale now on</p>
```

- (a) Write an additional JavaScript function to change the colour of Sale now on to black.

1

- (b) Re-write the paragraph element to include a call to your function from part (a), so that the text can also be changed to black when the mouse moves away.

2



\* X 8 1 6 7 6 0 1 2 6 \*

15. Some CSS properties for a website are implemented/coded as follows.

```
header {margin-right:10px; margin-left:10px;}
nav {margin-right:10px; margin-left:10px;}
#event {width:700px; margin-left:10px;}
#gaminglogo {width:75px; height: 75px; float:right;
margin-right: 20px;}
#eventlogo {float:right; width:75px; height:75px;
margin-right:20px;}
#floorplan {margin-left:10px; width:700px;}
footer {margin-left:10px; margin-right:10px;}
```

Using grouping selectors to remove any repetition, re-write the code to make it more efficient.

3

[Turn over



16. Music Tutor is a company that offers music lessons. They are creating a website to advertise the business.

A 'Contact Us' page will contain a form where customers supply their details and make an enquiry.

The HTML code for the form is shown below.

```
<form>
  Name: <br>
  <input type="text" name="name"><br><br>

  Email Address:<br>
  <input type="text" name="email"><br><br>

  Mobile Number:<br>
  <input type="text" name="mobile"><br><br>

  Instrument:<br>
  <select name="instrument" size="3">
    <option value="clarinet">Clarinet</option>
    <option value="flute">Flute</option>
    <option value="trumpet">Trumpet</option>
    <option value="tuba">Tuba</option>
    <option value="violin">Violin</option>
  </select>
  <br><br>

  Contact Method:<br>
  <input type="radio" name="contact"
  value="Phone">Phone Call<br>
  <input type="radio" name="contact"
  value="Email">Email<br>
  <input type="radio" name="contact" value="Text">Text
  Message<br>
  <br>

  Additional Comments:<br>
  <input type="textarea" name="comments" rows="10"
  cols="30">
  <br><br>

</form>
```



16. (continued)

- (a) (i) State which essential HTML form element is missing from the form. 1

\_\_\_\_\_

- (ii) Write the additional line of HTML code to add the missing form element. 1

- (b) (i) During usability testing of the form, feedback indicated that the preferred default contact method should be email.  
State the attribute that should be added to the email option to achieve this. 1

\_\_\_\_\_

\_\_\_\_\_

- (ii) A customer wants to select both violin and flute but the form does not allow this.  
Re-write the line of HTML code to allow this. 1

- (c) A range of personas are used when conducting usability testing of the website.  
State one suitable persona and explain why adopting this persona is helpful when conducting usability testing. 2

Persona \_\_\_\_\_

Explanation \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

[Turn over



16. (continued)

When designing the website the image and heading are to be on the same line as shown below.



The following HTML and CSS code is used.

```
<head>
  <style>
    img {float: right;}
  </style>
</head>

<body>
  <h1>Music tuition since 2006 </h1>
  
...

```

However, the image appears on the line beneath the heading, as shown below.



The following CSS rule is added.

```
h1 {display: inline;}

```

- (d) Explain why the display property of the `h1` element had to be changed to achieve this layout.

2

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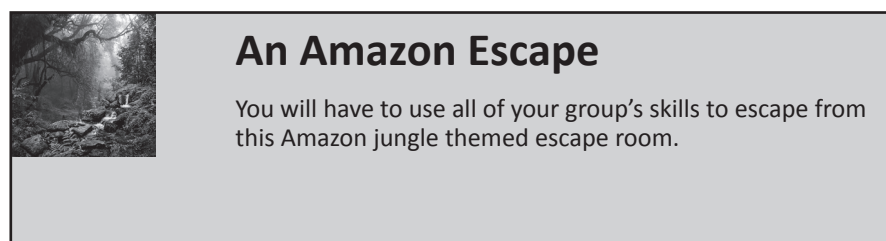
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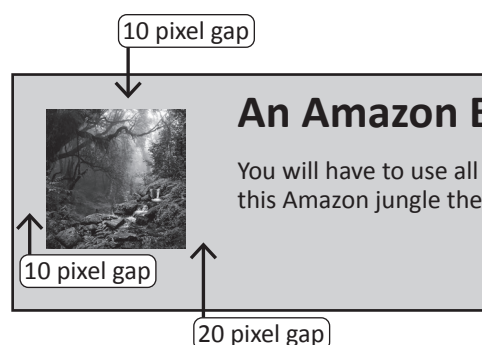
17. Pranav runs an escape room company that uses a website to promote the different themed escape rooms.

(a) Part of the information page of the company's website is shown below.

The image is too close to the top left of the section element.



The layout should be as below. The gaps above and to the left of the image are 10 pixels. The gap to the right is 20 pixels.



Complete the appropriate CSS rule for this image.

2

```
.eventimage {
```

[Turn over



17. (continued)

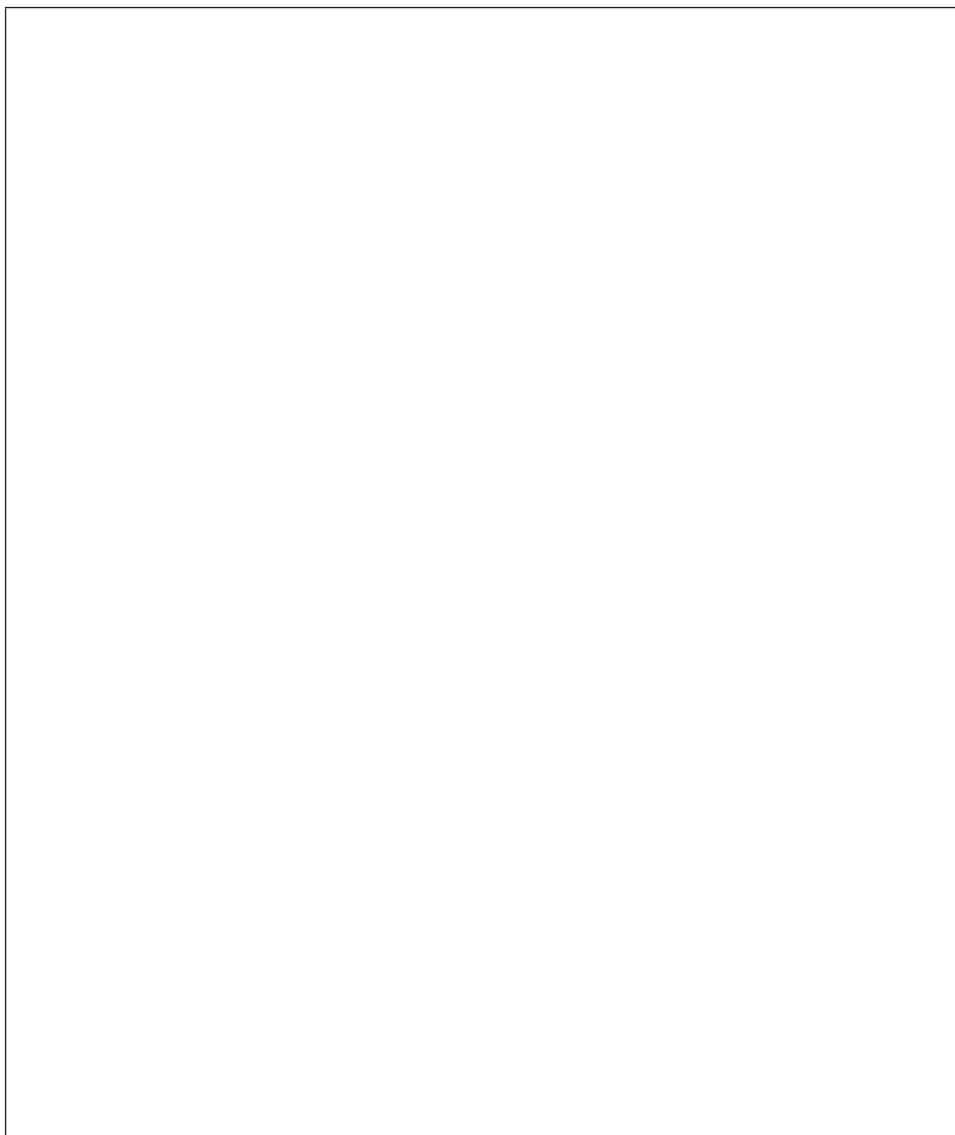
Pranav wants to add a new web page with a form where customers can contact his company to enquire about bookings.

Customers are required to enter a contact name, email address, date of booking and the party size (max of 6).

Customers will have to choose a theme from Amazon Escape, Mayan Mayhem or Aztec Anarchy. They can also provide any additional information.

(b) Draw a suitable wireframe for this form which indicates validation.

4




17. (continued)

- (c) The wireframe is then used to create a low-fidelity prototype.

Describe one benefit for the developer and one benefit for the client of using this low-fidelity prototype.

2

Developer benefit \_\_\_\_\_

\_\_\_\_\_

Client benefit \_\_\_\_\_

\_\_\_\_\_

- (d) The navigation bar for the website is as shown below.

• Amazon Escape • Mayan Mayhem • Aztec Anarchy • Contact Us

- (i) This version of the navigation bar used the code below.

```
nav {height:35px; background-color: darkgrey}
nav ul li {float:left;width:150px;text-align:center;}
nav ul li a {display:block; padding:8px;}
```

Complete the CSS rule to remove the bullet points from the navigation bar.

1

nav ul { \_\_\_\_\_ }

- (ii) Explain why the unordered list in the navigation bar will display differently from any other unordered list in the website.

2

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

[END OF SECTION 2]

[END OF QUESTION PAPER]



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